Week 6: Architectural design

Lab 6 — Chosen architecture: MVC architecture

Model–View–Controller (MVC), implemented as a small Flask application. MVC separates data and business rules (Model), user interface (View), and request handling (Controller). For a student-sized prototype, a modular monolith is simpler to test and deploy than microservices. The system can be split into services later if real usage and operational needs justify it.

Requirement	Design response
Security and privacy	Role checks on the server; ownership checks for student records; password hashing if real authentication is implemented; no sensitive data in logs.
Maintainability	Separate routes/controllers, service logic, data models/repository, templates, and tests.
Usability	Responsive pages, clear validation, status labels, and reference numbers.
Reliability	Database transactions, error handling, automated tests, version control.
Scalability	Start with a modular monolith and SQLite for the local prototype; use a managed production database only after requirements and deployment are approved.

Component/package diagram — recreate in draw.io

